

9<sup>th</sup> Oct  
2022

GRADIENT  
DECENT &

PERFORMANCE  
A MATRIX FOR  
REGRESSION

# Linear Regression Algorithm

~~\* (1) Simple~~

\* (1) simple Linear Regression

$$h_0(x) = \theta_0 + \theta_1 x$$

\* (2) Multi Linear Regression  
multiple

$$h_0(x) = \theta_0 + \theta_1 x_1 + \theta_2 x_2 + \theta_3 x_3 + \dots + \theta_n x_n$$

Here, with respect to multi Linear Regression my equation will be changing based on the number of independent features.  
How many independent features will be there, ( $\theta_n x_n$ ).

\* Simple equation linear regression about the equation of straight line.

## \* Conversation Algorithms :

Last Class, we were discussing about this 'Conversation Algorithms'.

'Conversation Algorithms' all about MSE (mean Squared Error). The (MSE) is about the 'Cost Function'.

How we was writing,

$$* J(\theta_0, \theta_1) = \frac{1}{m} \sum_{i=1}^m (h_{\theta}(x^{(i)}) - y^{(i)})^2$$

\* This is a Cost Function

\* This can also called as Loss Function

\* Difference between Cost Function vs Loss Function?

\* (i) 'Cost Function':-

In terms of Cost Function, we will be doing with respect to all the point.

$$\frac{1}{m} \sum_{i=1}^m$$

$i=1$  to  $m$

~~(M is a)~~

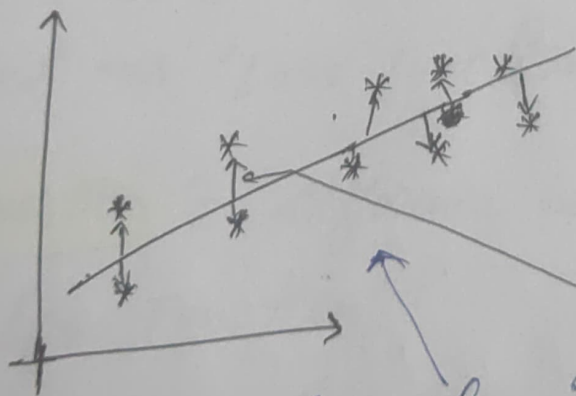
( $M$  = NO. of data points)

\* (ii) Loss Function :-

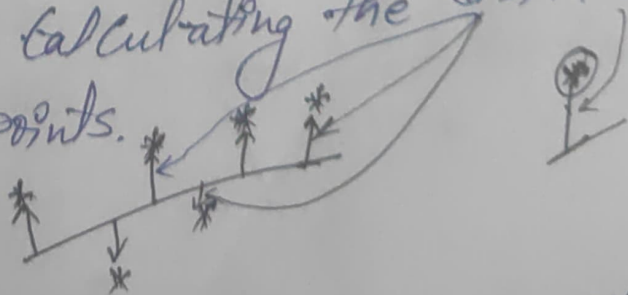
In case of 'Loss Function', it will be for every observation

for example :-

If I am a this data point.



which is like this, in 'Loss Function' (MSE) found out by calculating the distance between all these points.



But, if we talk about 'Loss Function', it with respect to each & every thing out there.

nothing left, everything.

\* Loss Function  $= (h_{\theta}(x^{(i)}) - y^{(i)})^2$

If I want to write Loss Function, this way.

This is a 'Loss Function'.

\* Loss Function is nothing but difference

between  $(\hat{y}^{(i)} - y^{(i)})$

predicted value

Actual value

or truth point

$(\hat{y}^{(i)} - y^{(i)})$  it is a difference

\* Explanation :- Revision what we  
Learn previous class

previous class, we learned about the

'Cost Function'

Equation was :-

\* 
$$J(\theta_0; \theta_1) = \frac{1}{m} \sum_{i=1}^m (h_{\theta}(x^{(i)}) - y^{(i)})^2$$

This is nothing but (MSE) - Mean Squared Error.  
This is basically a 'Cost Function'. we also  
called as 'Loss Function'.

\* Question :- Difference between 'Cost Function  
& Loss Function'?

Both are in the same only 50/50 left  
hand side & right hand side.  
But usually in ~~cost~~ 'Cost Function'.

For every point

the distance get  
calculated &



$$(h_{\theta}(x^{(i)}) - y^{(i)})^2$$

will do the average  $\rightarrow \frac{1}{m} \sum_{i=1}^m$

that is the region we say it as a MSE (mean Squared Error). & that is the ~~reg~~ region we divided by  $(1/m)$ .

\* Loss Function:-

But Case of 'Loss Function'.

If I calculate the distance between these 2 points also.



Then for every point, if I am calculating I usually say as a 'Loss Function'.

The 'Loss Function' equation will give like this

$$* \text{ Loss Function} = (h_0(x^{(i)}) - y^{(i)})^2$$

\* 'Loss Function' is about single point.

\* How we was discussing this part?  
when I used to create a line.



Let's say we have some data point, & let's say  
I created a 'best fit line'.

This is  $(h_0(x^{(i)}) - y^{(i)})^2$  nothing

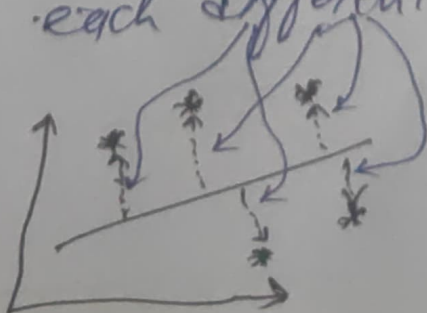
but we trying to find out the difference  
between all these points.

predicted & Actual points.

If I talk about 'Loss Function'

is for every point, we will try to calculate  
the distance.

'Loss Function' says that, I have to  
take care of each difference.



~~Loss~~

$$\text{Loss Function} = (h_0(x)^{(i)} - y^{(i)})^2$$

For every point, I will try to calculate the difference ~~Independ~~. Independently, for every single.

\* Cost Function applied for all the data points.

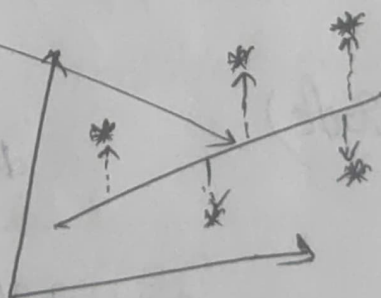
## \* Conversation Algorithms:

previously class we what we have discussed. what we used to do in conversation algorithms?

we used to update the  $(\theta_0, \theta_1)$  & we used to reduce this value.

$$\frac{1}{m} \cdot \sum_{i=1}^m (h_{\theta}(x^{(i)}) - y^{(i)})^2$$

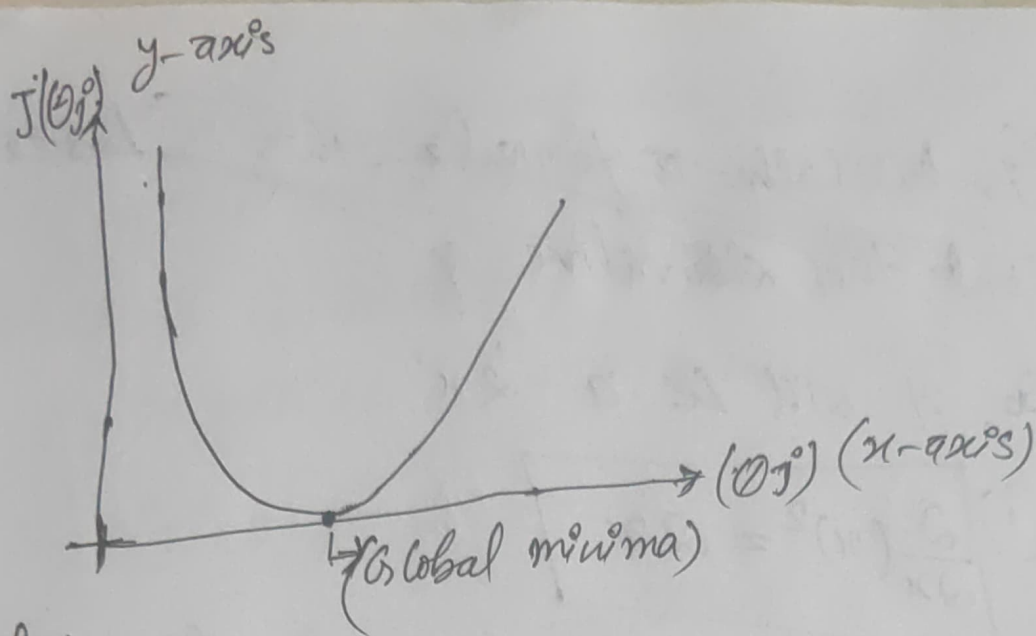
This value reduce only by changing this best fit line.



In 'Conversation Algorithm' we used to write ~~reg~~. Repeat until convergence.

$$\theta_j := \theta_j - \alpha \frac{\partial}{\partial \theta_j} J(\theta_j)$$





\* Notes:-

This is we already discussed in previous Lecture, so go & revised it.

Now, let's understand that,

what is the most important thing over here?

The ~~import~~ most important thing over here is to calculating this

$$\frac{\partial}{\partial \theta} J(\theta)$$

\* How to find derivative (2) mathematically?  
 what is the derivative of  $\frac{\partial}{\partial x} (x)^2$  with respect to derivative 'x'?

~~This is basically a formula, this is basically find out the derivative of.~~

So, it will be a  $2x$

So, (1)  $\boxed{\frac{d}{dx}(x)^2 = 2x}$

This is a basic thing; basic formula.

If we want to find out the

(2)  $\boxed{\frac{d}{dx}(x)^n = nx^{n-1} \cdot (1)} \leftarrow (x)$

derivative (2) of derivative (x) ~~of~~ 'x' to the power (n). with respect ~~to~~ derivative 'x'.  
then it will be

$nx^{n-1}$  then derivative of ('x') which is nothing but (1).

\* what will be  $\frac{d}{dx}(x+1)^2$ ?

(3)  $\boxed{\frac{d}{dx}(x+1)^2 = 2 \times (x+1) \cdot x(1+0) = 2(x+1)}$

It will be the ~~2~~ 2 multiply ~~x~~ by  $(x+1)$

then we try to find out the derivative of this entire term.  $(x+1)$ .

what is (2) derivative of 'x'

It will be  $(1) + 0$ . because this is a constant

So,  $(1+0)$  //

So, my final answer would be  
 $= 2(x+1)$  //



|                                                  |                                                                                  |
|--------------------------------------------------|----------------------------------------------------------------------------------|
| $\frac{\partial}{\partial \theta_j} J(\theta_j)$ | <del><math>\frac{\partial}{\partial \theta_j} J(\theta_0, \theta_1)</math></del> |
|--------------------------------------------------|----------------------------------------------------------------------------------|

I am going to find out the this particular part. this particular derivative (2).

$$\frac{\partial}{\partial \theta_j} J(\theta_0, \theta_1)$$

Here, 'J' value how many 'J' are there,  $\theta_0$  &  $\theta_1$  I have to compute. If I will write  $\theta_0, \theta_1$  & Let's say I have to find out the 1st derivative derivative (2) with respect to

$(J\theta_0) \rightarrow$   ~~$\frac{\partial}{\partial \theta_0} J(\theta_0, \theta_1)$~~   $\frac{\partial}{\partial \theta_0}$

$\partial \theta_0$

~~then~~ then how this equation will work,

$$\frac{\partial}{\partial \theta_0} J(\theta_0, \theta_1) = \frac{\partial}{\partial \theta_0} \left[ \frac{1}{m} \sum_{i=1}^m (h_{\theta}(x^{(i)}) - y^{(i)})^2 \right]$$

Explanation equation:

Now, if this derivative (2), I need to find out

$$\left[ \frac{\partial}{\partial \theta_j} \right]$$

this (J) value how many 'J' values are there, main  $\theta_0$  &  $\theta_1$  I have to compute, so I will write

$J(\theta_0, \theta_1)$  & Let's say, I want to find out 1st derivative (2) with respect to  $\theta_0$ ?

$$\partial [J_{\theta_0}] ?$$

If we are saying (2) derivative of  $\theta_0$  with respect to derivative (2) of  $J(\theta_0, \theta_1)$

then how this equation works?

$= \frac{\partial}{\partial \theta_0}$  in bracket  $\left[ \frac{1}{m} \right]$  what is my

'Cost function' this entire cost function  $J(\theta_0, \theta_1)$

it will be  $\sum_{i=1}^m$  (submission of  $i=1$  to  $m$ )

$(h_{\theta} x^{(i)} - y^{(i)})^2$  ( $h_{\theta}$  of  $x^{(i)}$  -  $y$  of  $(i)$  whole square)

\*Question:-

if we remember why we write

$\left[ \frac{1}{2m} \right]$ ? or  $\left[ \frac{1}{2} \right]$  why?

There was a reason why we have write,

Because, if I have to find out the

(2) derivative of this entire term

$$\left[ \frac{1}{2m} \cdot \sum_{i=1}^m (h_{\theta}(x)^{(i)} - y^{(i)})^2 \right]$$

with respect to  $\theta_0$  ~~and~~  $\theta = '0'$  for this

derivative (2) purpose only it used to write.

Because, if this square is there

$$\left[ \frac{1}{2m} \sum_{i=1}^m (h_{\theta}(x)^{(i)} - y^{(i)})^2 \right]$$

then it will come over here just ~~also~~ replace '1' So, it will become like this  $\left( \frac{2}{2m} \right)$  then

this  $\frac{2}{2}$  get cancelled.

$$* h_{\theta}(x) = \theta_0 + \theta_1 x$$

~~So~~

$$J = \theta_0 = \frac{\partial}{\partial \theta_0} \left[ \frac{1}{2m} \sum_{i=1}^m ((\theta_0 + \theta_1 x)^{(i)} - y^{(i)})^2 \right]$$

Explanation of above :-

what is this  $h_{\theta}(x)$  or  $h_{\theta}$  of  $x = ?$

$h_{\theta}$  of  $x$  is nothing but its a  $\theta_0 + \theta_1 x$ .

Now,

~~1st~~

Here, I am trying to calculate the  $\frac{\partial}{\partial \theta_0}$  derivatives of  $\theta_0$  with respect to

$$J = \theta_0$$

So, I will write  $\frac{\partial}{\partial \theta_0}$  derivatives of  $\theta_0$

$\frac{1}{2m}$  &  $\sum_{i=1}^m$  (Submission of  $i=1$  by  $m$ ).

I will replace this  $h_0(x)^{(i)}$  with

$$(\theta_0 + \theta_1 x)^{(i)}$$

So,

I will this way.

$$J = \theta_0 = \frac{\partial}{\partial \theta_0} \left[ \frac{1}{2m} \sum_{i=1}^m \left( (\theta_0 + \theta_1 x)^{(i)} - y^{(i)} \right)^2 \right]$$

Continue...  
next to next  
page

Next,

For example :

I am trying to find out derivative (2)  
of 'x' ?

~~$\frac{\partial}{\partial x} \frac{1}{2m}$~~   ~~$\frac{\partial}{\partial x} \frac{1}{2}$~~

$$\frac{\partial}{\partial x} \left[ \frac{1}{2m} \sum_{i=1}^m \left( (0_0 - 0_1 x)^{(i)} - y^{(i)} \right)^2 \right]$$

Now, I will write

$$\frac{\partial}{\partial x} \left[ \frac{1}{2m} \sum_{i=1}^m \left( \underline{(1+x)} - y^{(i)} \right)^2 \right]$$

I am replacing <sup>this</sup>  $(0_0 - 0_1 x)^{(i)}$  with this

On If I try to find out derivative (2)  
this  $[2]$  of square will come down, so it  
will become  $\frac{2}{2m}$  & 2 & 2 get cancelled

$$= \frac{2}{2m} \left[ (1+x) - y^{(i)} \right]$$

Now, if I will take this entire  $(1+x)$   
 & try to find out what it will become?  
 It will be multiply by  $\boxed{1}$

$$= \left[ (1+x) - y^{(i)} \right] \times 1$$

$$= \boxed{(1+x) - y^{(i)}}$$

Since, the it multiply by 1 doesn't change  
 the value, ~~the exp~~ it remain <sup>as</sup> <sub>same</sub> it is.

Now, Come back to our old equation

Continue.....

$$\boxed{J = 0_0}$$

$$* J = 0_0 = \frac{\partial}{\partial 0_0} \left[ \frac{1}{2m} \sum_{i=1}^m \left( (0_0 + 0_1 x)^{(i)} - y^{(i)} \right)^2 \right]$$

$$= \frac{\partial}{\partial 0_0} \left[ \frac{2}{2m} \sum_{i=1}^m \left( (0_0 + 0_1 x)^{(i)} - y^{(i)} \right) \right]$$

~~$$\frac{\partial}{\partial 0_0} \left[ \frac{1}{m} \sum_{i=1}^m \left( (0_0 + 0_1 x)^{(i)} - y^{(i)} \right) \right]$$~~

~~Here~~

$$= \frac{\partial}{\partial \theta_0} \left[ \frac{1}{m} \sum_{i=1}^m \left( \theta_0 + \theta_1 x^{(i)} - y^{(i)} \right) \right] \times \frac{1}{\rightarrow}$$

why multiply by  $\boxed{1}$ , because <sup>derivative of</sup>  $\theta_0$  is equal to  $\boxed{1}$ .

~~$$= \frac{\partial}{\partial \theta_0} \left[ \frac{1}{m} \sum_{i=1}^m \left( \theta_0 + \theta_1 x^{(i)} - y^{(i)} \right) \right]$$~~

$$= \frac{\partial}{\partial \theta_0} = \frac{1}{m} \sum_{i=1}^m \left( \theta_0 + \theta_1 x^{(i)} - y^{(i)} \right)$$

If we multiply by  $\cancel{1}$  ~~to this~~ it does not change the value, it remains ~~as~~ same as it is.

After doing the  $\partial$  derivative ~~of~~ with respect to the  $(\theta_0)$  we get this above equation.

\* Similarly, ~~if~~ if we continue with

$$\boxed{\theta_1} \quad \boxed{J_{\theta_1}} \quad \text{or} \quad \boxed{\theta_1}$$

Let's say we have to find out derivative (2) with respect to  $\boxed{J_{\theta_1}}$  or  $\boxed{\theta_1}$

then, the equation will work like this.

$$J_{\theta_1} = \frac{\partial}{\partial \theta_1} \left[ \frac{1}{2m} \sum_{i=1}^m \left( (\theta_0 + \theta_1 x^{(i)}) - y^{(i)} \right)^2 \right]$$

$$= \frac{\partial}{\partial \theta_1} \left[ \frac{2}{2m} \sum_{i=1}^m \left( (\theta_0 + \theta_1 x^{(i)}) - y^{(i)} \right) \right]$$

$$= \frac{\partial}{\partial \theta_1} = \frac{1}{m} \sum_{i=1}^m \left( \theta_0 + \theta_1 x - y^{(i)} \right) \times 1 \times x$$

why multiply by  $\boxed{x}$ , because derivative of  $\theta_1$  is equal to  $\boxed{x}$ .

$$J_{\theta_1} = \frac{\partial}{\partial \theta_1} = \frac{1}{m} \sum_{i=1}^m \left( \theta_0 + \theta_1 x^{(i)} - y^{(i)} \right) x^{(i)}$$

\*  $J_{\theta_0}$  = we multiply by (1) because  
it is a  $\theta_0 = 1$  &

\*  $J_{\theta_1}$  similarly we multiply by (x)  
because it is a value  $\theta_1 = x^{(i)}$

~~So, we did derivatives derivative (1) with  
respect to the  $J(\theta_1)$  or  $J(\theta_0, \theta_1)$~~

~~So, final outcome is this~~

\*  $J_{\theta_0}$   
 ~~$\theta_0$~~

So, we did derivatives (o) with respect to the  $J(\theta_0)$  or  $J(\theta_0, \theta_1)$

So, final outcome is this.

\* (1)  $J_{\theta_0}$

$$* J_{\theta_0} = \frac{\partial}{\partial \theta_0} = \frac{1}{m} \sum_{i=1}^m (\theta_0 + \theta_1 x^{(i)} - y^{(i)})$$

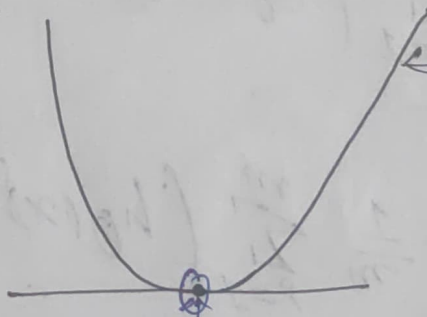
$$\theta_0 := \theta_0 - \alpha \frac{1}{m} \sum_{i=1}^m (h_{\theta}(x^{(i)}) - y^{(i)})$$

\* (2)  $J_{\theta_1}$

$$J_{\theta_1} = \frac{\partial}{\partial \theta_1} = \frac{1}{m} \sum_{i=1}^m (\theta_0 + \theta_1 x^{(i)} - y^{(i)}) x^{(i)}$$

$$\theta_1 := \theta_1 - \alpha \frac{1}{m} \sum_{i=1}^m (h_{\theta}(x^{(i)}) - y^{(i)}) x^{(i)}$$

\* Because of this update of equation in graph what we will get, it keep updating. But as of now we will get this graph in the right hand side. This kind of curve, called parabola ~~type~~ or 'Gradient descent curve.'



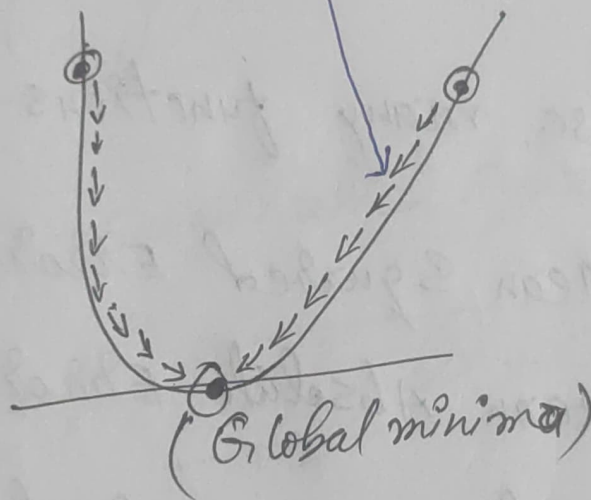
unless & until, if we don't come near to this region (Global minima), where our loss is minimum.

\*  $\theta_0$  is intercept.

\*  $\theta_1$  is slope.

then  ~~$\theta_2$~~   $\theta_2, \theta_3, \theta_4$  we can compute like this.

At the end of the ~~the~~ day, wherever our point would be, we will be going to this direction. because there is learning ~~rate~~ involved over here.



Q: what is the use of learning rate ( $\alpha$ )?  
• 'learning rate' is nothing but speed of conversation.

# Cost Function or Loss Function

\* How many Cost Function or Loss Function, we have?

we have these many functions available

- ① MSE : Mean Squared Error
- ② MAE : Mean Absolute Error
- ③ RMSE : Root Mean Squared Error.

(1) MSE :-

Formula :-

$$MSE = \sum_{i=1}^n \frac{(y - \hat{y})^2}{n}$$

$\hat{y} = 0_0 + 0_1 x$   
 $\uparrow$  predict value

or,

$$\left[ \frac{1}{n} \sum_{i=1}^n [(y - (0_0 + 0_1 x))^2] \right]$$

$\hat{y} = \hat{y}$  or  $\hat{y}$  (hat) mean over here  $0_0 + 0_1 x$

In the case of simple linear regression, or I can say this is my predicted value.

## Advantages & Disadvantages of MSE.

① The advantage of this particular equation. This is a Quadratic Equation

$$\frac{(\hat{y} - y^1)^2}{n}$$

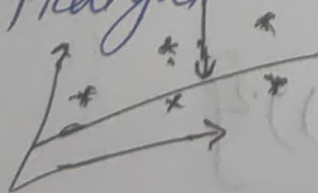
Q: what is Quadratic Equation?

Formula :-  $\boxed{ax^2 + bx + c = 0}$

$a, b, c$  = known numbers, where  $a \neq 0$

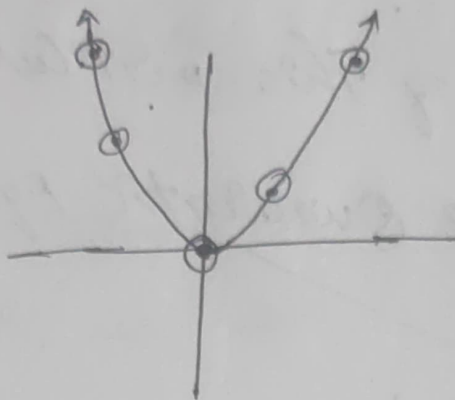
$x$  = the ~~unk~~ unknown

we know that our equation of the straight line is  $\boxed{y = mx + c}$



Q: what is the importance of Quadratic equation?

If we go & check graph.



If we check this we can see that we are getting ~~parabola~~ curve or Gradient Descent curve.

MSE Formula

$$MSE = \sum_{i=1}^n \frac{(y - \hat{y}_i)^2}{n}$$

← (Quadratic equation)

or

$$\frac{1}{n} \sum_{i=1}^n [(y - (\theta_0 + \theta_1 x))^2]$$

~~Handwritten scribbles and crossed-out equations, including  $(a-b)^2 = a^2 - 2ab + b^2$ .~~

~~Handwritten scribbles~~

$$ax^2 + bx + c = 0$$

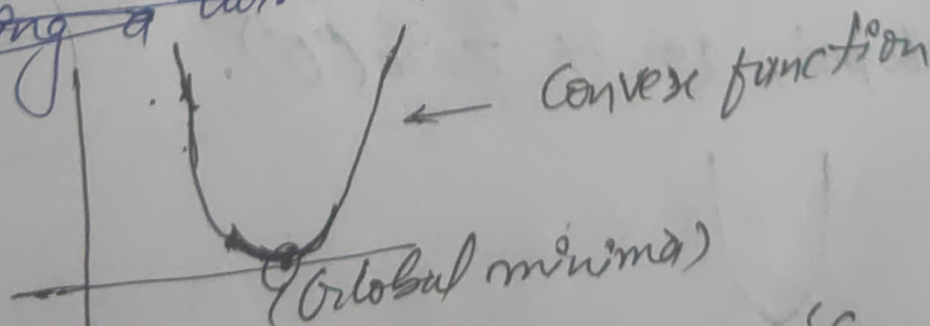
## Advantage - MSE ~~or advantage~~

① This equation is ~~differentiable~~ differentiable.  
why require  
differentiability to  
calculate the 00 & 01.

② If this equation  
differentiable then this  
equation has only one  
Global minima.

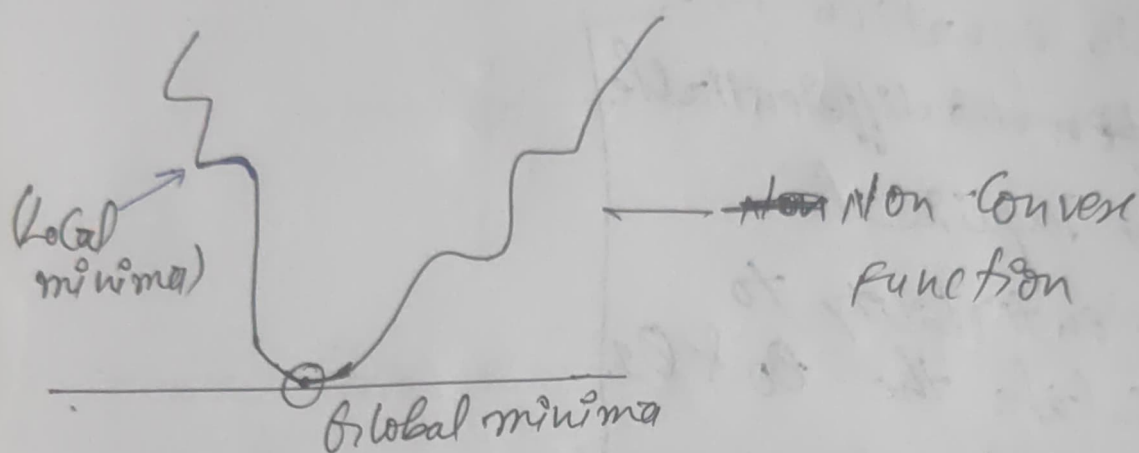
what does this  
mean?

Because of this I am getting this  
~~getting a curve~~ parabola curve.



we usually called this an 'Convex function'.

Similarly, if we take example of an another 'Convex Function.'



This is a ~~non~~ Non-Convex Function.

Both Convex & Non-Convex Function have the 'Global minima'.

Q: what difference between these 2 graph?

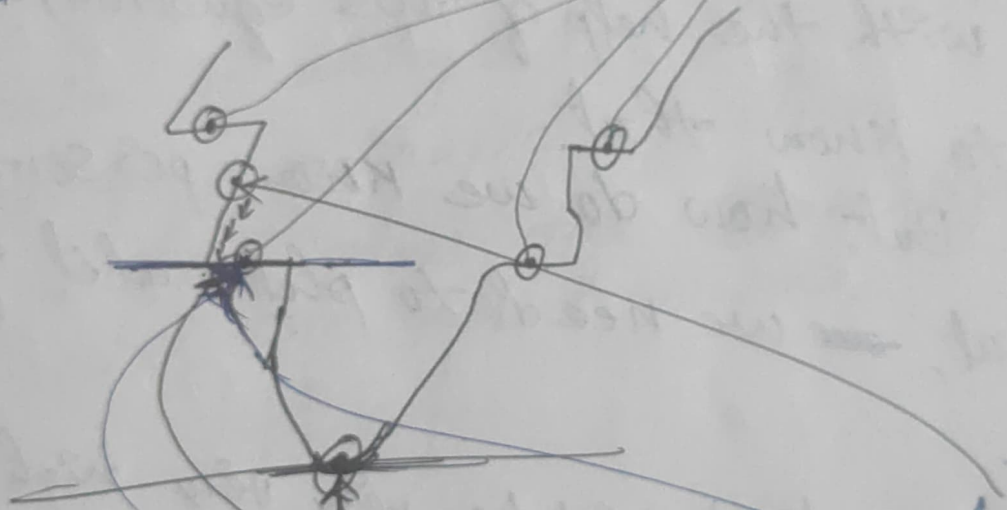
In 'Convex Function' if we try to find out the 'slope', is it zero ('0').



If it is zero ('0') my  $\theta_0$  or  $\theta_1$  will not get updated.

Now, In 'Non Convex Function'. what about  
Global minima, if I try to find out  
the slope's derivative?  
Is it zero?

But what about this point



Let's say my initial point is here, then  
it keep updating & come over here.  
Now since from here (0.0 & 0.1)

is not getting updated, because slope is  
zero. It will always get stuck over here.

It will think that, OK I have come to  
the 'Global minima'. But that is not  
the 'Global minima'. This point is 'Global',  
minima

\* This kind of 'Non Convex' scenario, will never come with this ~~particular~~ particular equation.

$$MSE = \sum_{i=1}^n \cdot \frac{(y - \hat{y})^2}{n}$$

~~How~~ with the help of this equation, we came to know that.

But, how do we know personally?

For that, ~~we~~ we need to plot a lot of things.

\* 'Loss Function' will be very very high.

\* 'Cost Function' will be very very high.

It will not keep on decreasing.

There may be very much scenario for that.

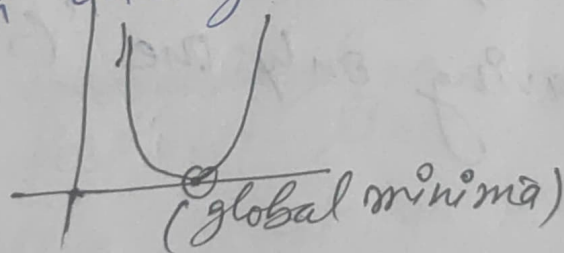
\* Non-Convex Function

For non-convex function, we have a lot of equation, but that equation is not used as a 'Cost Function'.

For Non-Convex Function, Next lecture  
~~Paraphrase~~ Logistic Regression algorithm  
we will discuss this more in detail.

There will be a 'Non-Convex'  
function because of that, we need to change  
that 'Non-Convex' to 'Convex Function'.

\* Our aim is always to work with this  
kind of Quadratic equation.  
which gives the ~~paraphrase~~ parabola  
curve with one global minima.

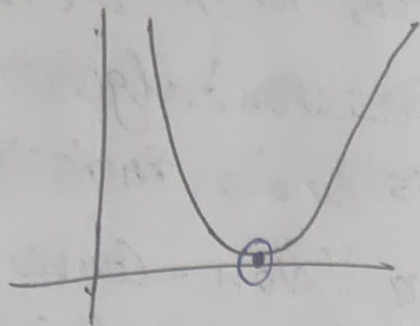


That is done by the MSE (mean squared error).

\* M.S.E

Formula is: 
$$\sum_{i=1}^n \frac{(y - \hat{y})^2}{n}$$

This formula is a quadratic equation.  
If whenever we plot a 'Quadratic  
equation' we always get this kind of



~~parabola~~ Curve  
para bola

para para Curve we get.

so we find any 'Local minima'  
like we found inside 'Non Convex Function'  
Curve.

No right, ~~it has only one~~

So, Convex Function means

we will be having only one 'Global  
minima'.

\* But, in Non-Convex Function, we may  
have many 'Local minima'.

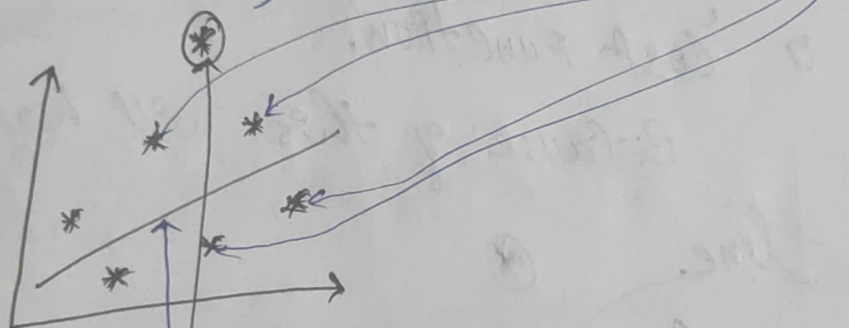
## Dis Advantage - MSE

- ①. ~~This~~ This equation is not robust to outliers.

what does this mean:-

Let's say,

I have this data points



I have a ~~straight line~~ Best fit line. After creating best fit line, I got an outliers, which is like this.

Because of this outlier whether my 'Cost function' will increase or decrease? It will increase by tremendous

amount.

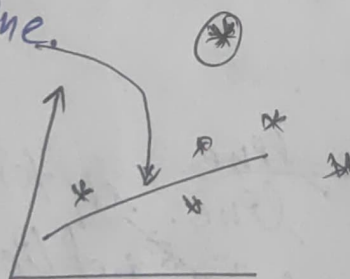
If my 'Cost Function' is increasing then, don't we think my algorithms go ahead & update the  $[00 \& 01]$  in case of simple linear regression.

if it is updating the formula is in such a way that the difference, I am trying to square it up

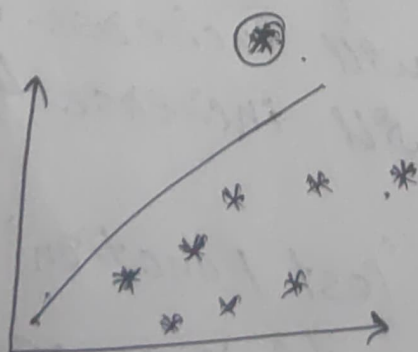
$$MSE = \sum_{i=1}^n \frac{(y - \hat{y}^1)^2}{n} \quad \text{or} \quad \frac{1}{n} \sum_{i=1}^n [y - (\theta_0 + \theta_1 x)]^2$$

this is a 'Cost Function'.

Because of this will happen to this line.



This line will get updated by a very big margin.



So, it jump or make a big switch, but this should not have happened.

Because of this outlier. Since, we are



~~penalizing~~ penalizing the this error.

$$\frac{1}{n} \sum_{i=1}^n [(y_i - (0_0 + 0_1 x_i))^2]$$

~~Error~~ Error whenever there, we squaring it up.

So, Because of this ~~squaring~~ squaring we can see the difference, ~~a~~ a huge jump. what we need to do is that,

In order to solve this problem, we need to remove the outliers.

So, from this equation (MSE) if we directly use it, it will not robust to outliers, because there is a biggest jump or switch.

## Ops Advantage - MSE

(2)

$$(y - \hat{y})^2$$

Now in 'Cost Function', I am trying to calculate  $(y - \hat{y})^2$  or,  $y - y_{\text{hat}}(1)$  whole square (2)

Let's say, we have a 'Data': Experience | Salary

| Exp | Salary (Lakh)<br>INR |
|-----|----------------------|
|     |                      |

Exp = Independent features

Salary = Dependent features

In dependent features, let's say 'Salary' is in Lakh (INR).

If I am trying to calculate  $(y - \hat{y})^2$

means,  $(\overset{\text{(Lakh)}}{\text{Salary}} - \overset{\text{(Lakh)}}{\text{predicted salary}})^2$

whatever difference is there we are squaring this

in lakhs

(Lakh)<sup>2</sup>

$$\frac{\text{Salary}}{(\text{Lakh})} - \text{predicted-salary} = (\text{Lakh})^2 //$$

If we calculating it, we are getting final unit is (Lakh)<sup>2</sup>.

Is the unit changing!  
(Lakhs)<sup>2</sup>

Yes, right, because Lakh + Lakh = (Lakh)<sup>2</sup>  
unit is changing.

from Lakh, it now (Lakhs)<sup>2</sup>  
we are now going to focus more into (Lakh)<sup>2</sup>.

Based on that, we have to reduce the 'Cost Function'.

\* This is the Error that is present  
$$\left( \frac{\text{Salary}}{(\text{Lakh})} - \text{predicted-salary} \right)$$

~~that is present~~  
But this unit is (Lakh)<sup>2</sup>

$$* (y - \hat{y})$$

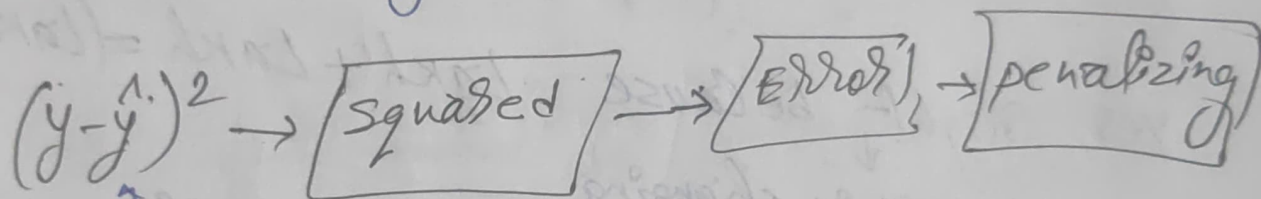
My 'cost function' is  $y - \hat{y}$  at (1)

my unit is in (Lakhs) only.

My Error unit was (Lakhs)

only.

Error getting squared over here.



The main purpose over here is that,

this 'Error' over here is getting squared.

when, we say 'Error' getting squared.

that basically means that Error, is getting penalized.

penalizing basically means we

are increasing the Error.

If the Error is coming still we are increasing it.

54'23

For Example 2:-

This is a real value &

$\boxed{100001}$

this is a predicted value

$\boxed{100002}$

$$(100001 - 100002)^2 = ?$$

I am trying to find out the Error.

then it will be  $\boxed{1}$

$$* (100001 - 100002)^2 = \boxed{1} \quad \leftarrow$$

\* Suppose, this is our predicted & this our Error.

$$(100 - 110)^2 = ?$$

what is squaring part?

It will be the  $\boxed{100}$

$$(100 - 110)^2 = \boxed{100} \quad \leftarrow$$

we just thinking with respect to one variable.

$$\boxed{(y - \hat{y})^2} \rightarrow \uparrow \uparrow \uparrow$$

Because of this 'Cost Function',  
it increasing the Error.

For Example 13:

Formula:

$$MSE = \sum_{i=1}^n \frac{(y - \hat{y})^2}{n}$$

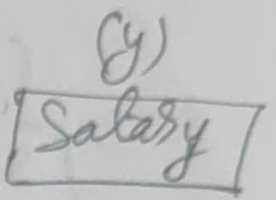
this is my 'Cost Function', I need to reduce it  
↓↓↓, unless I untill, I don't reduce it, it  
will not work.

Let's say, Exp is one of my variable & another  
variable 'Salary'

| <u>Exp</u> | <u>Salary</u> |
|------------|---------------|
| -          | -             |
| -          | -             |
| -          | -             |

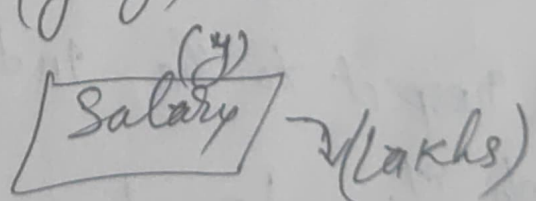
~~Salary~~ Salary ← dependent features

\* we are trying to understand what is the disadvantage of MSE?

So, this is my 'y' 

I am creating various model it is working perfectly fine.

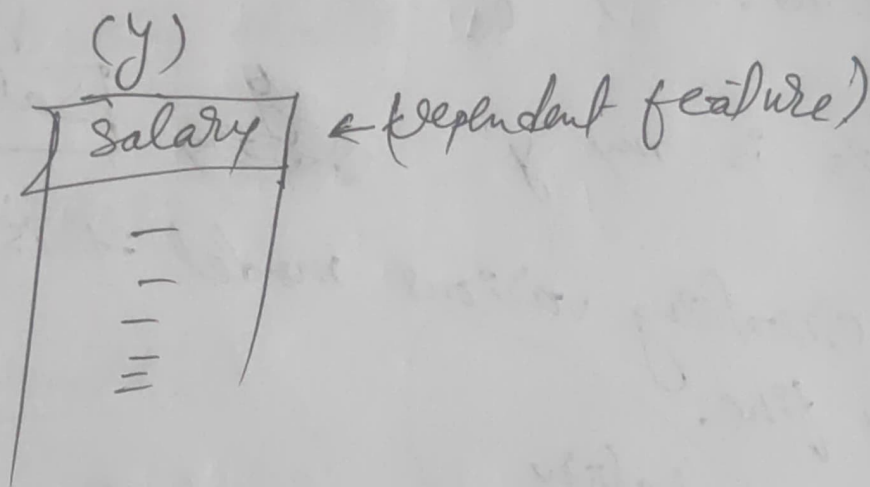
EXP      salary  
—      —  
—      —  
—      —  
—      —

But, when I do  $(y - \hat{y})^2$ , initially the unit was in (Lakhs) 

Now, I am using it as  $(\text{Lakhs})^2$ , I am changing the unit, because of that what is happening, Time Complexity will increase. I am changing the unit all together, because of these outlier  of this will also happen.

Note:-

~~we~~ we can't or we don't do  
Standardized for the output features.

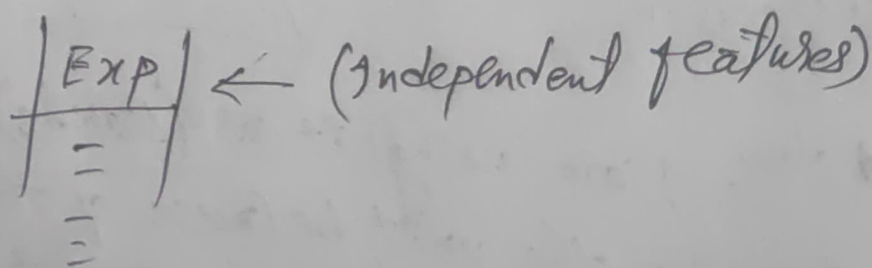


we ~~can't~~ don't do Standardized for Errors

$$(y - \hat{y})^2$$

→ (Lakhs)

\* with respect to the independent features only  
we do it.



\* If we are penalizing the Error that basically means, we are changing the unit in short.

Because of that ~~it will both~~  
~~happen~~, it will not be robust to outliers.

\* To solve this problem, we remove the outliers. For that, we have many techniques.

---

(2)

(2) MAE :- Mean Absolute Error

Formula :-

~~$$MAE = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i|$$~~

$$MAE = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i|$$

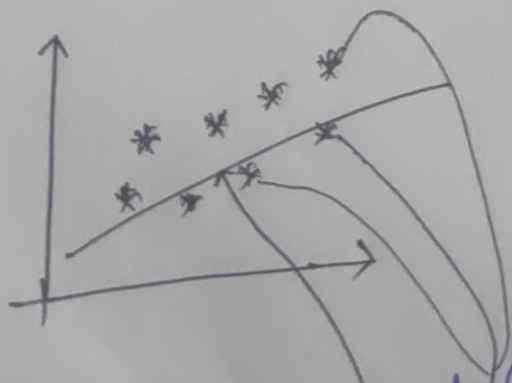
$y_i$  = prediction

$x_i$  = true value

$n$  = total number of data points

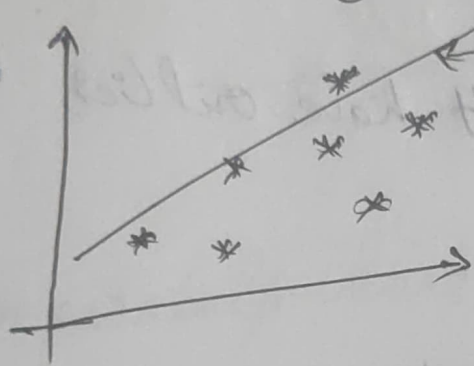
Advantage

(i) Not penalizing the error, Can we say this is robust to outliers.



Let's say, I have some data points like this  
this is my best fit line.

what will happen, if I get the outlier?  
 this line will change, but with the  
 smaller margin like this, not bigger jump  
 small jump.



It will not get attracted toward outliers.  
 because we are not squaring ~~it~~ it.

(ii) whatever advantage in MSE, here also  
 it will be ~~same~~ in the same unit.

$$|y - \hat{y}|$$

we will not get negative this is an  
 absolute value.

It means, even though, I will do

Subtraction  $|5 - 8| = |-3| = 3$

$$\boxed{5 - 8 = (-3)}$$

whenever; we do absolute value, it will be always a positive number |3|

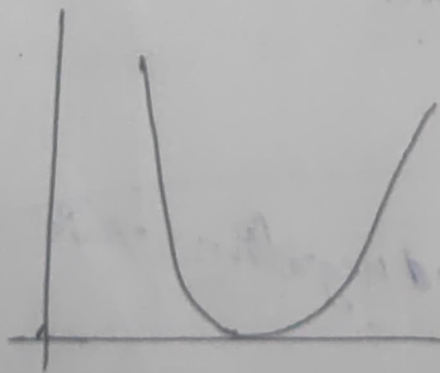
(b) Disadvantage:

\* we don't use MSE?

No, ~~we~~ if we don't have outliers, then ~~don't~~ use MSE.

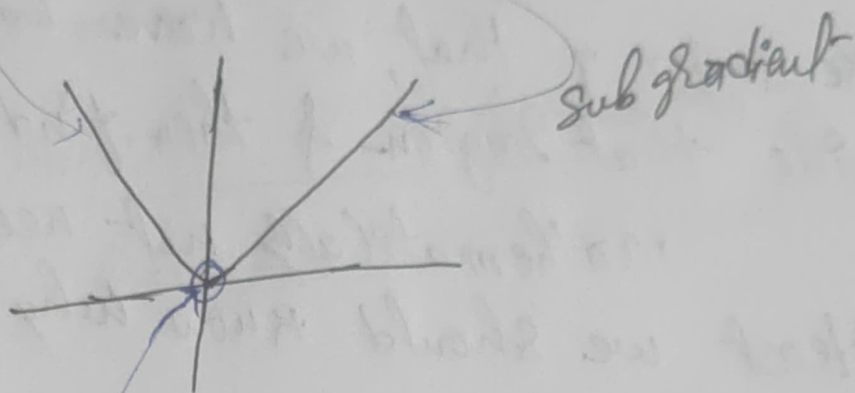
(b) DisAdvantage: (MAE)

(i) Convergence usually takes more ~~than~~ time, optimization is a complex task.



In case of MSE, we are getting this kind of parabola curve, because it is a quadratic equation.

But, Here this is not Quadratic equation,  
the lines comes. Something like this



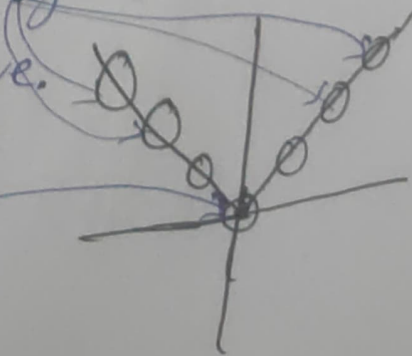
~~parabola~~ parabola looks something like this,  
this 2 lines converging at the origin.

The disadvantage is that, we can't  
find out the derivative, when our point  
is zero.

In this case, how the derivative is  
basically calculated.

we basically have to use  
sub-gradient concept.

Sub gradient says that, we have to  
~~div~~ divide region by region & try to  
find out the derivative.  
To come up near  
to this



we have to basically find out part by part sub-gradient.

Sub-gradient means, we have to take region  $g$  that, we have basically divide that region  $f$  then find it out.

Mathematically not needed but at least we should know why convergence taking time.

~~(i) Time Consuming~~

(ii) Time Consuming

---

~~\* Loss Function or Cost Function~~

## \* Simple Linear Regression

we know about so far :-

\* Loss Function or Cost Function :-

what all loss Function we used.

① MSE

② MAE

③ ~~RMSE~~

④ Huber Loss (in sep learning we have deeply explained)

(4) RMSE

(5) performance matrix

Next,

\* performance matrix

Task to solve or work

① Huber Loss

② RMSE



## PERFORMANCE MATRICS

- ① R squared Error
- ② Adjusted R squared
- ③ Overfitting & underfitting (Bias & variance)

## \* performance Metrics :-

How do we check model is good or not in the case of simple linear regression?

For that, we have to go ahead & see the 'performance metrics.'

In performance metrics, we see 2 things

① R Squared Error

② Adjusted R Squared

1.18.20

\* Adjusted R Squared & R Squared Error, actually help us to check the performance of the model.

## (1) R-Squared Error

With respect to R-Squared,

Let's say I have these data points



Formula :-

$$R \text{ squared} = 1 - \frac{SS_{Res}}{SS_{Total}}$$

\* what is  $SS_{Res}$  ?

It is nothing but Sum of Square Residuals.

$SS_{Res} = \text{Sum of Square Residuals.}$

\* what is  $SS_{Total}$  ?

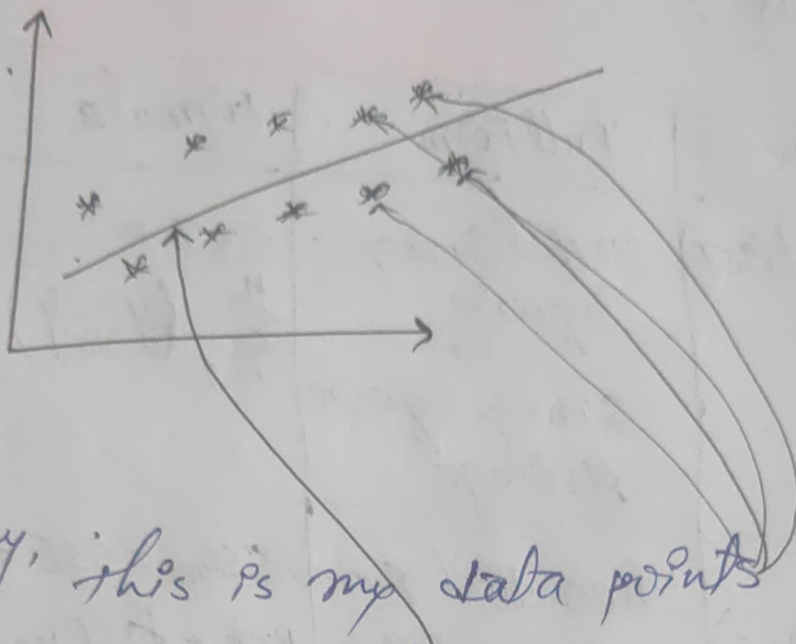
It is also nothing but Sum of Square Average. or just Sum of Square TOTAL Sum of Squares.

## Formulas :-

| Name               | Full Form                                           | Formula                                             |
|--------------------|-----------------------------------------------------|-----------------------------------------------------|
| SSTotal (SST)      | Total Sum of Squares<br>or<br>Sum of Square Average | $SST = \sum_{i=1}^n (y_i - \bar{y})^2$              |
| SSres<br>(SSE/RSS) | Residual Sum of Squares                             | $SS_{res} = SSE = \sum_{i=1}^n (y_i - \hat{y}_i)^2$ |

description

- ① Total variation of observed values from their mean
- ② variation of observed values from model predicted values (unexplained by the model)



Let's say, this is my data points  
 this is my best fit line.

So, Formula:

$$R\text{-Squared} = 1 - \frac{SS_{Res}}{SS_{Total}}$$

$$= 1 - \frac{\sum_{i=1}^n (y_i - \hat{y}_i)^2}{\sum_{i=1}^n (y_i - \bar{y})^2}$$

$\rightarrow$  low value  
 $\rightarrow$  High value

$$R\text{ Squared} = 1 - \left\{ \frac{\text{Small number}}{\text{Bigger number}} \right\} \rightarrow \text{Small number}$$

$$= \leq 1$$

$$= 0.85 \quad \text{or} \quad 0.75$$

$$= 85\% \quad \text{or} \quad 75\%$$

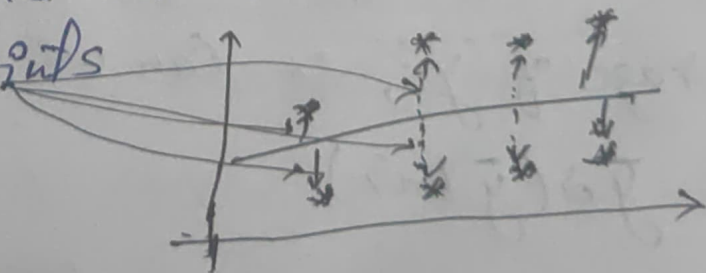
### Formula Explanation:-

$$\text{So, } 1 - \frac{SS_{Res}}{SS_{Total}}$$

So, SS Res, if I go ahead & write, it will be written as  $1 - \sum_{i=1}^n (y_i - \hat{y}_i)^2$

1 - Submission of  $i=1$  to  $n$ ,  $y_i - \hat{y}_i$  (1) hat whole square.

In short what we are doing here, mean square only we are doing, we are just going to find out Error between the particular points



& we are calculating the distance.

Now,

Below. I am going to write SS Total

It will be written as

$$1 - \sum_{i=1}^n (y_i - \bar{y})^2$$

1 - Submission of total  $i=1$  of  $n$ ,  $y$  of  $(i) - \bar{y}$   $y$  with cross or bar over here.

$\bar{y}$  : It is nothing but average of  $y$ .

what is the average of  $y$ ?



we can write this line, this will be my average of  $y$ , so, can we say this will be  $\bar{y}$  ( $\bar{y}$  bar)

\* So, now model is fitted well, then which will be a bigger upper one or lower one?

$$1 - \frac{\sum_{i=1}^n (y_i - \hat{y}_i)^2}{\sum_{i=1}^n (y_i - \bar{y})^2} \leftarrow \begin{matrix} \text{Lower value} \\ \text{High value} \end{matrix}$$

So, usually numerator ~~value~~ will be low value right. numerator means refers to the top part of a fraction or upper case.

So, my  $(SS_{Res})$  will be low value & my  $(SS_{Total})$  will be high value.

Next,

$$1 - \left\{ \frac{\text{small number}}{\text{Bigger number}} \right\} \leftarrow \text{small number}$$

can we write  $1 - \frac{\text{small number}}{\text{Bigger number}}$  divided by

if my model is fitted well.  
once we do this division my outcome will be of this division will also a small number.

So, what total R squared output I am always going to get.

I am always going to get less than 1  
 $\leq 1$  //

Whatever value, we are getting we can convert this into the percentage(%).

Let's say, this R squared get = 0.85, then

Can we say this is 85% accurate.

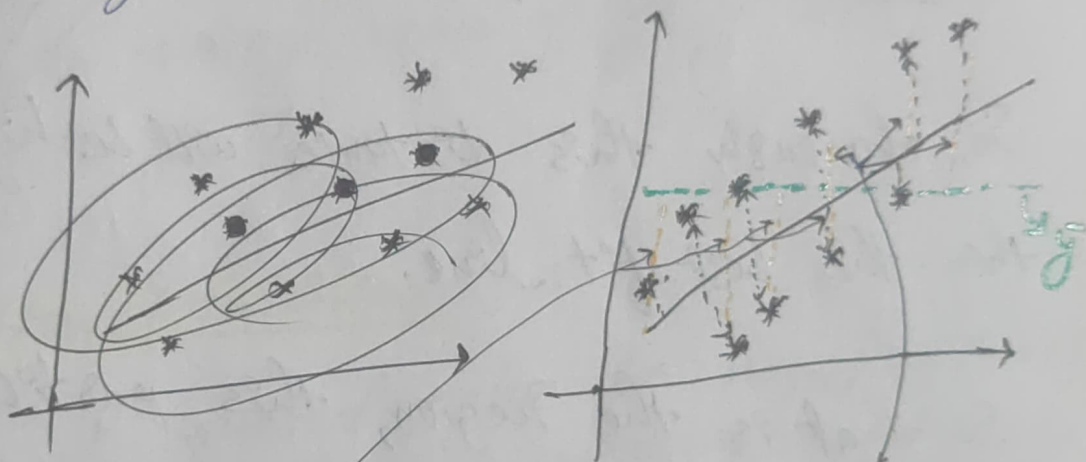
If it coming as 0.75, then

Can we say this is 75% accurate. this model.

Question ?

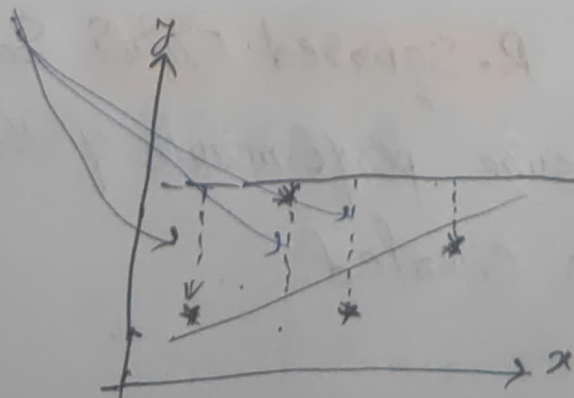
why how denominator is bigger  
(lower case)  
than numerator ?  
(upper case)

A . Denominator (lower case) will be  
bigger right why :-



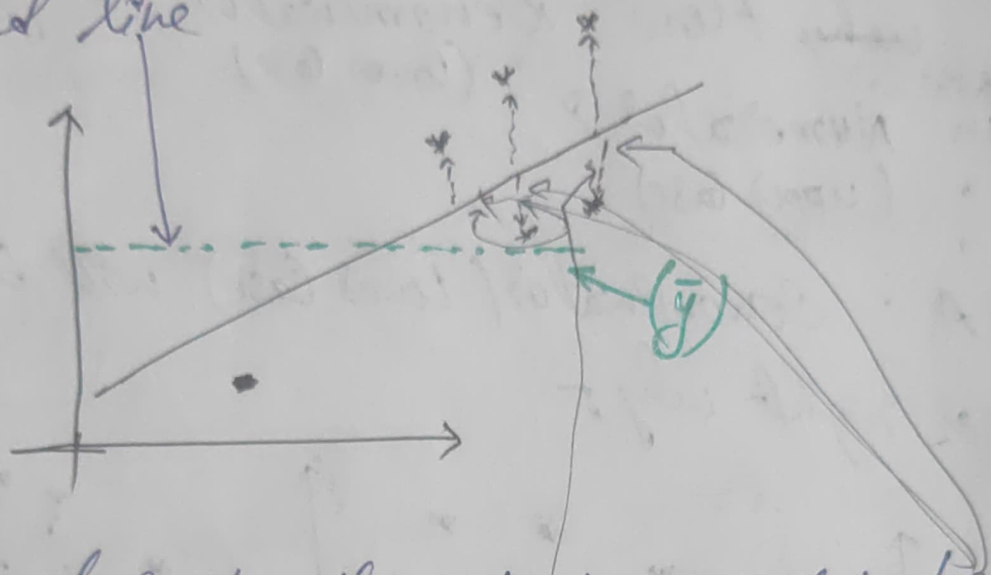
This Error, we will going to compute will  
always be less than this all Errors. ~~get~~

\* This all all Errors, because this is not ~~fitted~~



fitted perfectly fitted lines.

obviously then this will be not perfectly fitted line



So, obviously this distance will be higher than the best fit line.

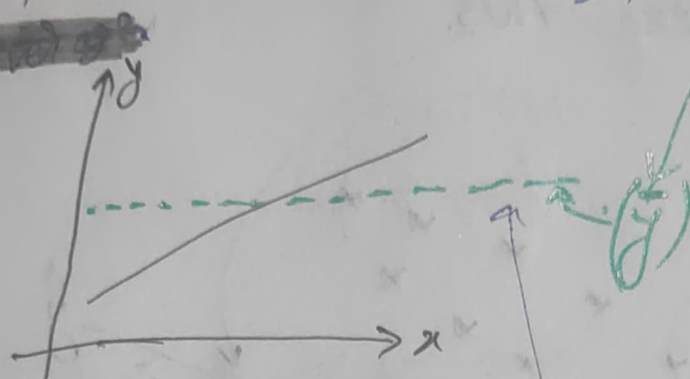
So that is the reason, this particular value denominator (lower case) will be higher.

$$\sum_{i=1}^n (y_i - \bar{y})^2 \leftarrow \text{higher value}$$

\* Note:-

\* This is what R-Squared Error says, it going to measure performance of the model that we have created.

\* Can we say that, this  $(\bar{y})$  I have created this is the basic model, ~~just~~



Just an average value of  $\bar{y}$  & it is trying to fit it.

### \* Question ? Interview

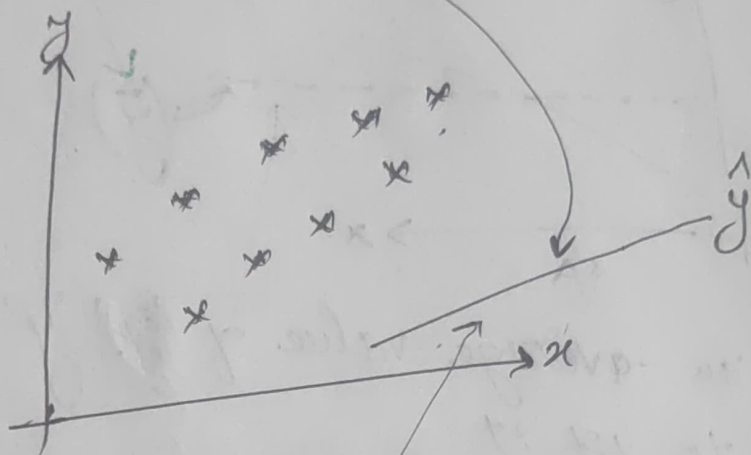
Can  $R^2$  squared value be negative (-ve)?  
 $R^2$  squared value  $\geq -ve$ . what we are going to do?  
 If it comes (-ve) negative the model is very bad, bad than this average line also.  
 So, imagine how bad it will be.

For example:-

Let's say scenario,  
 this are my data points &



Let's say, my model <sup>has</sup> fitted this line  
 something like this.



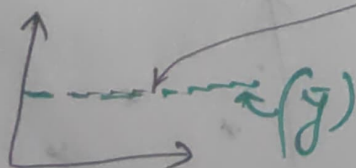
very bad model, worst in the life. see this.

\* So, just think this  $(\bar{y})$  model will  
 play better role or  
 this line will play?

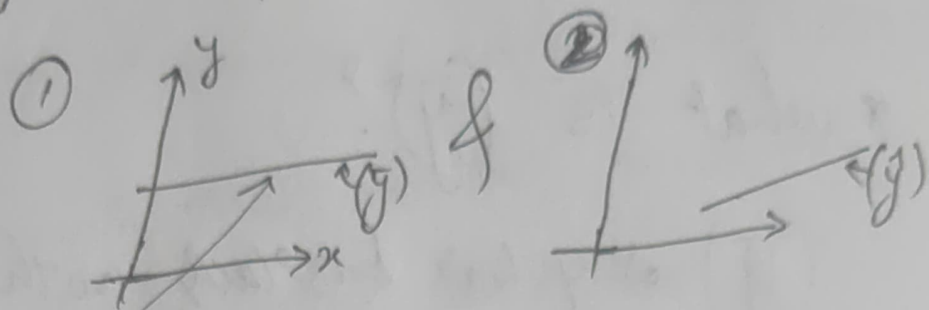


\* If we compare this 2 lines?

⊙ If I consider this is my best fit line  
 & this is my  $(\bar{y})$  or  $y$  bar.



~~Let's~~ if we compare this 2 lines?



This line will be much better, if we compare this 2 lines.

In this case what will happen, ~~my  $R^2$  value will be~~

~~my~~  $R^2 = (-ve)$

my  $R^2$  value will be negative (-ve).

So, this line will be  $(\bar{y})$  or  $y$  bar.

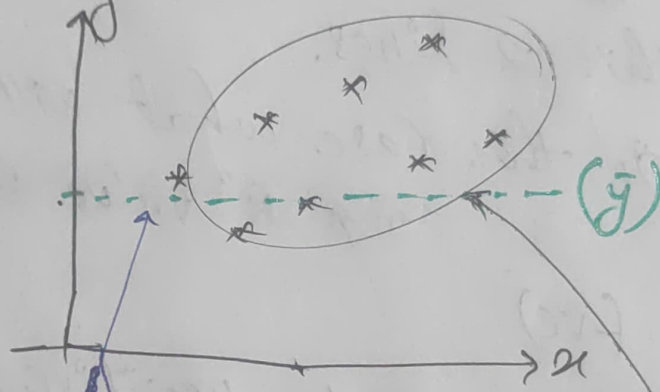
\* This line will be  $(\hat{y})$  or  $y$  hat.

\* Conclusion  $\hat{y}$  will be better than the  $(\bar{y})$ .

\* Question 1.

\* what is  $(\bar{y})$ ?

$(\bar{y})$  or  $y$  bar basically nothing average of this  $y$ 's.



what is average of all this points, somewhere will get center point  $(\bar{y})$ .

~~Q1~~

Question 5

$R^2$  coming what does this mean?

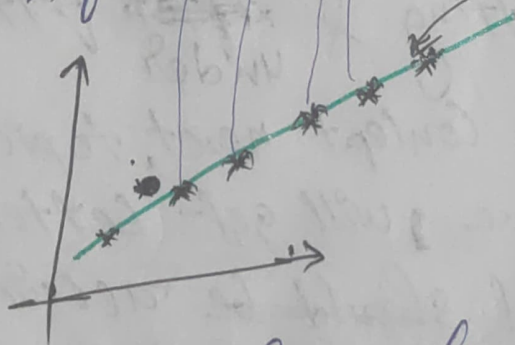
$R^2$  coming basically means,

\*Question 2:-

$R^2$  if it coming 1 what that mean?

$R^2$ , if it coming that basically mean that,  
for this particular point, ~~let's say,~~

Let's say,  
all the points in straight  
line & if it falls



then,  
I get a best fit line then I can get the

$$R^2 = 1.$$

(2)

## Adjusted R Squared



what is the ideal  $R^2$  we should get. more better right.

we will going to learn a concept called as

{ over fitting & ~~under~~ fitting }

just after this concept next topic.

Then we will get better picture what my model ~~should be~~ accuracy should be and all.

~~Let~~ Let's start :-

Let's say,

I have a feature which is called as size of the house. & I have my ~~dependant~~ feature which is price of the house.

|                                        |                                         |                                       |
|----------------------------------------|-----------------------------------------|---------------------------------------|
| Size of House<br>(Independent feature) | City; Location<br>(Independent feature) | Price of House<br>(Dependent feature) |
|----------------------------------------|-----------------------------------------|---------------------------------------|

Now,  
with respect to this particular ~~on~~ left hand side (size of house) feature & with respect to the right hand side (price of house) feature.

\* Do we get some kind of  $R^2$  value?  
yes, we get the  $R^2$  value.

\* Let's say,  
I am adding one more independent feature or parameter is [city location]

Because of this adding of feature, whether the  $R^2$  increase?

Yes, right there are ~~high~~ high chance  
to get an increase.

Because based on the (city location)  
my (price of house) also go up and down.  
I will be able to get more info  
information.

So, my ( $R^2$ ) will definitely increase.

Let's say,  
if I add one more new  
independent feature No. of bedrooms.

| Size of House | City Location | No. of Bedrooms | Price of House |
|---------------|---------------|-----------------|----------------|
|---------------|---------------|-----------------|----------------|

will my ~~feature~~ ( $R^2$ ) may increase,  
will I get a better model, which will  
be able to do a prediction of the  
price of house.

Yes, it will increase ~~that~~ we will  
get a better model & because of which  
I will be able to predict the price of house.

\* what if I will add one more feature.  
(~~Salary of the owner~~) Gender.

| Size of house | City location | No. of Bedrooms | <del>Salary of the owner</del> Gender | Price of house |
|---------------|---------------|-----------------|---------------------------------------|----------------|
|               |               |                 |                                       |                |

what now, if I try to calculate the  
(price of house) will the model increase  
or decrease? After adding one more  
feature 'Gender' who is staying in this  
house? will the model performed well  
or not?

No, right why because there is no  
correlation between these 2 features—  
Continue...

$\boxed{\text{Gender}}$  &  $\boxed{\text{price of House}}$

But, when we try to calculate

the  $\boxed{R^2}$

Based on this particular formula  
~~that is~~ whatever formula we have  
discussed so far.

we will be seeing that,

Let's say,

During ~~size of the~~  $\boxed{\text{size of the House}}$

&  $\boxed{\text{price of the House}}$  these 2 feature.

I was getting 65%. Let's assumed.

$R^2$  65%

$R^2$  75%

Then, I add one more feature  $\boxed{\text{City location}}$

Then suddenly the accuracy increase to 75%  
Let's assumed.

then, again I ~~add~~ add one more feature  
No. of bedrooms, then the accuracy increase  
to 88%. Let's assume.

$R^2$  . 65%

$R^2$  75%

$R^2$  88%

$R^2$  90%

then, again I try to add one more feature  
my accuracy increase to 90%.

Gender my accuracy increase to 90%  
when we try to add more feature I  
try to calculate the  $R^2$ . my  
accuracy increases. ~~to~~  
↓  
keep

we can see some pattern over here, usually  
this happens, whenever the features are  
very highly correlated each other, the  
increase in the accuracy will happen very  
quickly.

But, later on the accuracy will be very less increase.

$R^2$

88%

$R^2$

90%

why because there is no ~~correlation~~ correlation,  
but since we are able to calculate the  
 $\hat{y}$  or  $y$  hat & we are able to apply for  $R^2$ .  
we will be see there will be a  
very slight increase.

\* So we think this ~~slight~~ increase should  
happend.

$R^2$

88%

$R^2$

90%

No, ~~right~~ because it should not  
had happend.

Because these 12 features not

Gender & price of the house no correlation

between them.

In order to solve this problem,  
we used the Adjusted R Squared

Formula :-

Adjusted R Squared :

$$\text{Adjusted } R^2 = 1 - \frac{(1 - R^2)(N - 1)}{N - P - 1}$$

Here,

$N$  = No. of data points

$P$  = No. of independent features.

now, in this particular case,

when I say  $P = 1$  what does this mean,

It means, I have one independent feature Size of House.

when I had one independent feature.

~~my independent feature~~

my Adjusted  $R^2$ , if this is 65%.

|             |                                    |
|-------------|------------------------------------|
| $R^2$ . 65% | Adjusted $R^2 = 63\%$<br>( $p=1$ ) |
| $R^2$ 75%   | Adjusted $R^2 = 73\%$<br>( $p=2$ ) |
| $R^2$ 88%   | Adjusted $R^2 = 80\%$<br>( $p=3$ ) |
| $R^2$ 90%   | Adjusted $R^2 = 85\%$<br>( $p=4$ ) |

then this will be slightly less than  $R^2$ .  
So, it will be Let's say 63%.

this will be relation between  $R^2$  &  
Adjusted  $R^2$ .

why because the formula is like  
this only.

As we can see in formula of  
Adjusted  $R^2$ .

$$1 - \frac{(1 - R^2) \cdot (N - 1)}{N - p - 1}$$

As we can see over here, whatever  $R^2$ , I am

getting getting, I am subtracting  $\frac{(1-R^2)(n-1)}{n-p-1}$

the relation, we can see that, based on  $R^2$ , it will always be less than  $R^2$ .

next,

Let's say, another feature got added

City location

& this is highly

Co related.

Here

Adjusted  $R^2 = 73\%$ , & p value is 2

( $p=2$ )

next, it goes like this. next, ...

No. of Bedrooms ...

next,

But, when Gender feature is added, this scenario, what will happen is that the Adjusted  $R^2$  will not increase to 90%, instead it will decrease to 85%.

Adjusted  $R^2 = 85\%$  //

why because this feature Gender & price of the house  
both are not correlated.

\* This is a difference between the  $R^2$  & Adjusted  $R^2$ .

\* we will do practical also, when we keep adding some unnecessary columns, we will be seeing that, my  $R^2$  will be increase by smaller amount but in this Adjusted  $R^2$  it will not increase instead it will decrease.

\* Adjusted  $R^2$  will not make any impact on the accuracy if the unnecessary features is added.

Adjusted  $R^2$  will going to evaluate our accuracy based on very important features.

whom to believe, at the end of the day  
we should believe in Adjusted  $R^2$ .

\* whenever we have outliers, which one  
we should use, which loss function?  
= MAE (Mean Absolute Error)

\* if we don't have an outliers then what  
loss function we should use?  
= MSE (Mean Squared Error)

### (3) Overfitting & underfitting

are basically called it as (Bias & variance)

Let's say,

I have a specific dataset.

DATASET - 1000 datapoints

DATASET ← (this is my entire dataset)

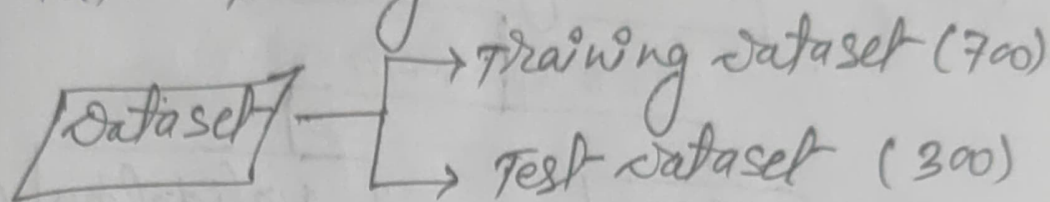
| Exp | years |
|-----|-------|
| -   | -     |
| -   | -     |
| -   | -     |
| -   | -     |
| -   | -     |
| -   | -     |
| -   | -     |
| -   | -     |
| -   | -     |
| -   | -     |

I have ~~data~~ around 1,000 data points.  
What is data points?

Let's say, I have features like Exp | years  
These are my data points →

Like this I have 1,000 data points.

Now, as soon we get this dataset, what we do is that, we divide this dataset into 'Training dataset' & Test dataset.



Let's say, out of 1,000 dataset (700) gone for Training dataset & (300) gone for Test dataset.

\* (1) Training dataset (700) - (Model Training)

The training dataset which is 700, actually being used for training model.

\* (2) Test dataset (300) - (Model Test)

The ~~entire~~ test dataset which is 300,

~~will be tested with this new dataset.~~  
the entire model will be ~~be~~ tested by this new dataset, which is called test dataset. This is considered as a complete new dataset to be tested.

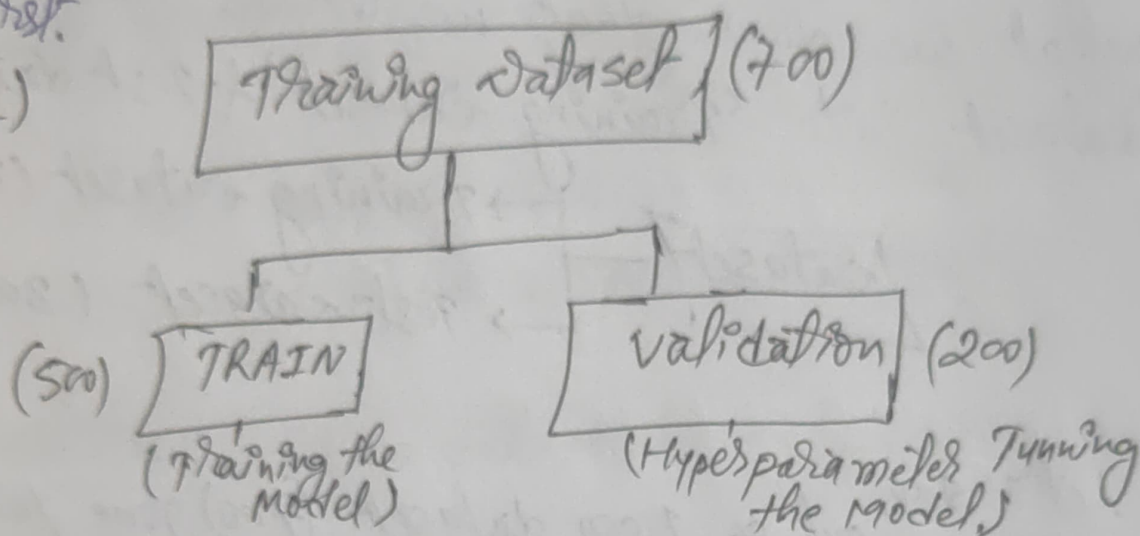
Note: This model has no idea about this ~~particular~~ dataset.

~~(1) Training~~

Let's take this 'Training dataset'

first.

(1)



This training dataset will be divided into 2 more parts. Training & validation data.

Let's say,

There are total (700 records)

(500 records) gone for training &

(200 records) gone for validation.

\* This (500 records) use for training the model &

\* This (200 records) remaining one ~~use~~ used for Hyperparameter Tunning the model.

We need to tune the model for that we need some validation data.

# Model

TRAIN DATA : very good Accuracy (90%) <sup>[Bias]</sup>

TEST DATA : very Good Accuracy (85%) <sup>[Variance]</sup>

Generalized model

So, let's say,

for my 'Training data' & for my 'test data'.

I have created the model & I have trained the model.

Let's say,

for this particular model with respect to the 'Training data':

I got a very good accuracy of (90%).

For Test data: also very good accuracy of (85%).

Is this accuracy good both of them?

Yes, obviously it's a good accuracy in both.

\* Our main aim should be, our model should be able to get good <sup>Training</sup> accuracy & good 'test accuracy'.

this kind of accuracy create  
a ~~good~~ generalised model.

~~whenever we get good accuracy  
with the 'Training dataset'.~~

\* usually 'Training data' information is  
given by a ['Bias']

\* usually 'Test data' information is  
given by a ['variance']

\* whenever we talk about 'Training data'  
accuracy, we are basically going to use  
'Bias'.

\* whenever we talk about ['Test data']  
accuracy, we are going to use ['variance'].

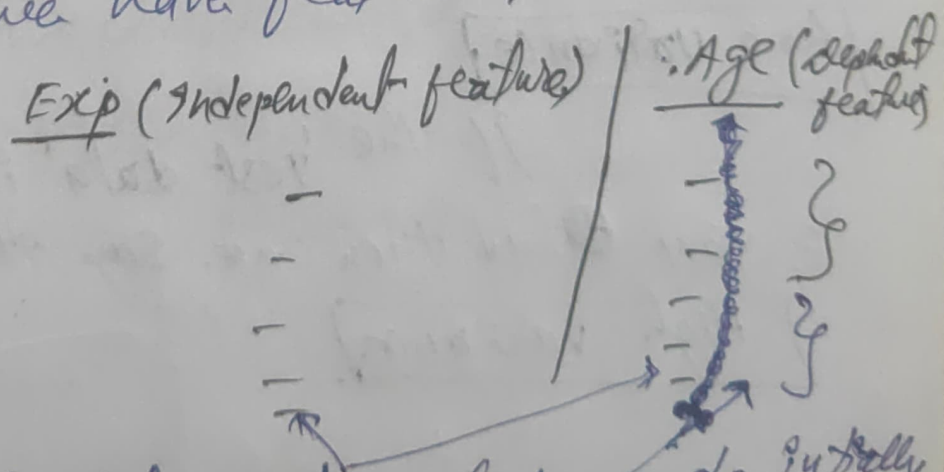
\* How Train data & Test data accuracy will be difference?

\* Model when it is giving the prediction for the 'Trained data' we will be get the Some accuracy.

\* For the 'Test data' also, we will be able to get some accuracy.

Because, in 'Test data' what will happen is that,

Let's say, we have features like Exp & Age.



These are our data points, what we do initially we divide the data into the 'Train' & 'Test data'.

Here, we know the truth value that is how we have to compare they have got the good accuracy or not?

\* Train data :-

If the 'train data' is very good accuracy, then we say it as a

'Low Bias'.

If the 'train data' is accuracy is very bad then we say it as a

'High Bias'.

\* Test data :-

If the 'test data' is very ~~good~~ good accuracy, then we say it as a

'Low Variance'.

If the 'test data' is accuracy is very ~~bad~~ bad then we say it as a

'High variance'.

## \* Scenario base Question ? [PART-1]

\* Train data : very good accuracy [90%] - [Low Bias]

\* Test data : Bad accuracy [50%] [High variance]

⇓  
[overfitting]

### \* Explanation above part:

\* Train:- Let's say 'Train data', we have very good accuracy, the model is giving to me. ~~as~~ [90%]. In this case, it will be 'Low Bias'.

\* Test:- For 'Test data', it is having a bad accuracy, let's say, it has (50%).  
In this case, it will be 'High variance'.

So, with respect to the 'Train data', I am getting a good accuracy. &  
with respect to the 'Test data', I am getting bad accuracy.  
Over this mean, this condition is called as 'overfitting'.  
basically

overfitting means the model has  
recified all the training data. It ~~can~~ bihated  
all the training data. Bihate in school,  
we bihate all the answer. right.

we all know Table from 2 to 12  
& Someone suddly asked what is  
 $12 \times 5 = ?$  don't know we ~~we~~ get shock.

Then we say  ~~$12 \times 5 = 60$~~ .

~~Suddenly they will not say, because  
machine did not learn that way.~~

Belause, ~~we have~~ they have recified  
in some manner.

Someone ask what is  $12 \times 5 = ?$ , then  
we start like

|                    |
|--------------------|
| $12 \times 1 = 12$ |
| $12 \times 2 = 24$ |
| $\vdots$           |
| $12 \times 5 = 60$ |

~~like this manner, suddenly machine ~~at~~  
can't say~~

Like this manner, ~~but~~ we  
can say  $12 \times 5 = 60$  <sup>directly</sup>, but in the machine  
Case they can't say, they will this pattern.

overfitting means train the model has  
biased ~~the model~~ all the  
information like this.

But, this should not happen in  
machine learning.

This condition is basically called  
as overfitting.

\* So, how do we solve this problem?  
we need to perform a 'hyper-parameter  
tuning'.  
or we need to increase the data.  
we will do \* in practical class.

## \* Scenario Based Question? PART-2

- \* Training data: For the training data, The Model accuracy is Low.
- \* Test data: For the test data, the model accuracy ~~is~~ <sup>it can be</sup> Low or High.
- \* Train data: Model accuracy is low [High Bias]
- \* Test data: Model accuracy it can be ~~High~~ Low or High. [Low or High variance]

⇓  
under fitting

So, in this model is 'under fitting'

under fitting mean, they can't even ~~be~~ rectified. \*

For example:

Some people don't study but when they give exam, choose the correct answer ~~pick~~ tick this & that, still got a good ~~score~~ score. tick Jai mata di.

This ~~case~~ ~~model~~ ~~is~~ ~~an~~ ~~underfitted~~ ~~model~~ but he performed well ~~with~~ ~~the~~ in exams. something like that.

### \* Generalized Model

In case generalized model, what will happen.

Let's say,

this is my data points, a training data.

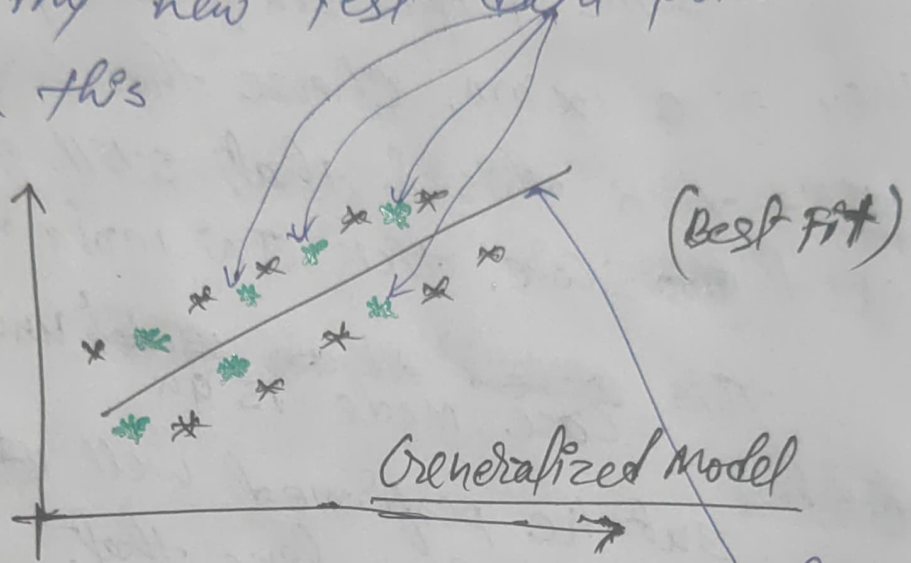


this is my best fit line.

This is my ~~set of training~~ Training data

Let's say,

If my new test data points  
comes like this



Don't we think ~~we~~ this ~~best fit~~ line  
will perform well.

Obviously it will perform well.

This scenario, we will get generalized  
Model. we will get in this case. This generalized  
Model is basically 'Best Fit.'

\* Scenario Question?

what will be in this ~~scenario~~  
scenario.

Let's say,  
This is my 'training data'. I get  
my best fit line, like this.



Suddenly, I get my 'Test Data,' which is  
like this.  
This is called the 'overfitting'

This is trained well with the 'training  
data.' But, with the 'test data,' it is not  
trained well.

Next,

Now Here this is my 'Training data' &



(Under fitting)

this is my 'test data' & this is my  
best fit line.

This scenario called as

'under fitting'

Note:-

This is all about the ~~linear~~

~~Linear~~ Regression